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Subject code :- UQ25CA653A

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Q1 Select all student in the CS department
→ $\sigma_{(Dept) = 'CS'}$ (Student table)

Q2 Project only the name and Dept of students

Q3 Select courses offered in 2024.

Q4 Project only the title of courses (no duplicates).

Q5 Find the union of student enrolled in in DBMS (C1) and OS (C5).

Q6 Find the intersection of students enrolled in DBMS (C1) and AI (C3)

Q7 Find students enrolled in DBMS (C1) but not in (OS) (C5).

Q8 Compute the cartesian product student × courses.

Q9 Select all enrollments where course in DBMS (C1)

Q10 Project all distinct Dept values from students.

Q1 $\sigma_{(Dept) = 'CS'}$ (Student table)

Rollno	name	Age	Dept
101	Alice	20	CS
103	carol	19	CS
106	Emma	21	CS

Q2 $\Pi_{\text{title}, (Dept)}$ (~~course~~ Student table)

~~title, Dept~~

~~DBMS~~

~~networks~~

name	Dept
Alice	CS
Bob	IT
carol	CS
David	ECE
emmy	CS

Q3 $\sigma_{year=2024}$ (course table)

courseID	title	Dept	year
C1	DBMS	CS	2024
C3	AI	IT	2024
C5	OS	CS	2024

Q4 $\sigma_{(Distence\ title)}$ (course table)

~~course~~ title
DBMS
networks
AI
mathematics
OS

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Date______ Page______

Q5 π ~~(Rollno (σ courseid = 'C1' (enrollment))~~

student table (Rollno, name) U Enrollment (cid)

Rollno	name	course id
101	Alice	C1
102	Bob	C1
102	Bob	C5
104	David	C1
105	emmy	C5

Q7

$\pi_{Rollno}(\sigma_{courseid = C1}(enrollment))$ —

$\pi_{Rollno}(\sigma_{courseid = C4}(enrollment))$

Roll no
101
102
104

Date____ Page____

Q.6 (student $\bowtie$ (Roll no, name) Π enrollment (cid))

Roll no	name	cid
101	Alice	C1, C3

Q-7 $\Pi_{Roll\ no}(\sigma_{courseid='C1'}(enrollment) - \sigma_{courseid='C5'}(enrollment))$

Roll no
101
104

Q-8

Roll no	name	age	Dept	courseID	title	Dept	year
101	Alice	20	CS	C1	DBMS	CS	2024
101	Alice	20	CS	C2	networks	CS	2023
101	Alice	20	CS	C3	AI	IT	2025
101	Alice	20	CS	C4	mathematics	ECE	2023
101	Alice	20	CS	C5	OS	CS	2024
102	Bob	22	IT	C1	DBMS	CS	2023
102	Bob	22	IT	C2	networks	CS	2024
102	Bob	22	IT	C3	AI	IT	2023
102	Bob	22	IT	C4	mathematics	ECE	2024
102	Bob	22	IT	C5	OS	CS	2023
103	carol	19	CS	C1	DBMS	CS	2024
103	carol	19	CS	C2	networks	CS	2023
103	carol	19	CS	C3	AI	IT	2024
103	carol	19	CS	C4	mathematics	ECE	2023
103	carol	19	CS	C5	OS	CS	2024
104	David	23	ECE	C1	DBMS	CS	2023
104	David	23	ECE	C2	networks	CS	2023
104	David	23	ECE	C3	AI	IT	2024
104	David	23	ECE	C4	mathematic	ECE	2023

Date______ Page______

Rollno	name	age	Dept	courseID	title	Dept	year
104	David	23	ECE	CS	OS	CS	2026
105	Emmu	21	CS	C1	DBMS	CS	2024
105	Emmu	21	CS	C2	networks	CS	2023
105	Emmu	21	CS	C3	AI	IT	2026
105	Emmu	21	CS	C4	mathematics	ECE	2023
105	Emmu	21	CS	C5	OS	CS	2027

Q-9 $\sigma_{courseID = 'C1'}$ (enrollment)

Rollno	courseID
101	C1
102	C1
104	C1

Q10

π (Distinct Dept) (student)

Dept
CS
IT
ECE